

Parents, Children, and ChatGPT in Home Environments: The Conversation Content and the Interaction Mode

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Abstract

AI-powered conversational tools (e.g., ChatGPT) have been increasingly adopted by families and incorporated into their daily conversations. This qualitative interview study investigated five parents of children aged 6 and 13 regarding the content and modes of their interactions with ChatGPT at home. The findings revealed that families primarily use ChatGPT for knowledge-driven and play-driven conversations with two primary interaction modes: low-moderation (parents allow children to independently interact with the tool) and high-moderation (parents actively guide and scaffold the interactions). The study highlights limited and age-related parental concerns about children's interactions with AI tools, offering insights into how these tools are integrated into daily home conversations and providing guidance for the development of child-centered AI technologies in home settings.

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1 Introduction

The emergence of artificial intelligence (AI) tools has transformed family interactions with technology, creating new opportunities for children's learning, play, and everyday problem-solving [36]. Conversational AI tools like ChatGPT emerged as intriguing resources for families, engaging parents and children in meaningful conversations and offering unique educational opportunities to support child development in home settings [8, 18, 24]. Understanding how AI tools are integrated into the everyday lives of children and families, as well as the roles parents play in mediating children's interactions with AI, is essential for identifying both the opportunities and challenges associated with its use. This highlights the need to explore the dynamics of parent-child interactions with AI in natural home environments. While prior research has extensively explored the role of AI in structured environments such as

classrooms or laboratories [14, 33, 38], the organic adoption of AI at home—specifically what it is used for and how it is used in the wild—remains underexplored [27]. Understanding how children and parents navigate these interactions in everyday life is crucial for guiding the responsible development and implementation of child-centered AI technologies.

This study aims to bridge the gap by exploring how parents and children navigate AI tools in their everyday lives, focusing on families who have adopted AI tools, such as ChatGPT, with young children at home. Through semi-structured interviews with five parents, the research examines the content and modes of interaction between children and ChatGPT, seeks to provide a nuanced understanding of conversational AI tool's role in home learning environments. Additionally, it seeks to inform the development of AI technologies that align with the needs of children and families, fostering meaningful interactions. Specifically, the study aims to address the following research questions:

- RQ1: What content do children and parents use ChatGPT for in daily conversations?
- RQ2: How do parents and children interact with ChatGPT in daily conversations?

2 Related Work

2.1 AI for Childhood Education

In recent years, research on AI tools such as voice-based conversational agents (e.g., Amazon Alexa and Google Home) and LLM-powered applications (e.g., ChatGPT) has been increasingly investigated for their utility in language and literacy learning such as storytelling, reading and writing [7, 10, 16, 17, 21, 22, 30, 35, 38], and learning programming languages [13, 19] and second languages [2, 10, 15, 31, 34, 35]. Education and AI researchers have published empirical studies and guidelines on AI integration for K-12 curricula, with various standards developed for K-12 classrooms to improve AI literacy [4, 12, 18, 21, 23, 29, 38]. Long et al. [18] further defined "AI literacy" as "a set of competencies that enable individuals to critically evaluate AI technologies; communicate and collaborate effectively with AI; and use AI as a tool online, at home, and in the workplace." Despite the increased attention on the educational use of AI tools, research on how parents and children utilize AI tools in the home environment has been limited [29, 36]. This is likely due to the practical limitations in gathering and collecting in-the-wild information about parent-child dyads' naturalistic interaction in the home, and the diverse parenting styles and beliefs in AI-mediated technology use. A majority of these studies have

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been conducted in controlled settings (e.g., schools) and lack explorations in natural settings such as home. Prior work found that early exposure to AI technology can help children better adapt to the concept of and ethical use of AI [4, 5, 28, 32]. These empirical studies motivated the present study to continue investigating the content and use of AI tools in the home setting.

2.2 AI Use in Parents and Children at Home

Parent-child joint media engagement is common in children during middle childhood (aged 6–11) and has been increasingly examined in recent research [8, 37]. Prior research has explored the use of various AI applications (e.g., voice assistant, ChatGPT) for daily tasks and has reported diverse use cases in child-AI interaction [1, 8], yet limited studies have examined the independent and collaborative use of AI tools by parents and their young children. For example, Druga et al. [8] conducted a five-week online in-home study with 18 children (5 to 11 years old) and 16 parents using AI tasks and found parent plays multiple roles (e.g., mediator, mentor, teacher, collaborator) that can significantly impact children's home media use and AI literacy. Since AI tools can empower parents with insights and resources to better support children's language and literacy development in the home setting, investigation on these different roles help HCI researchers understand how AI can provide personalized learning and high-quality learning resources to families from diverse backgrounds [29]. Additionally, as numerous AI tools offer interactive language features in various languages, they can be effective in supporting parents who may not have extensive knowledge or resources to aid their children's language learning [21]. However, research on AI literacy has not extensively examined parental roles, mediation, or concerns about AI, particularly given the diverse developmental trajectories across different age groups. To date, there are limited opportunities for researchers to investigate individualized practice of AI use in the home, and without ongoing studies examining in-the-wild AI tool use, HCI scholars cannot systematically identify AI use across developmental stages in younger children. By investigating these dynamics of AI use at home, the present study seeks to uncover children's natural interactions with AI technologies, while also highlighting the critical roles parents play in facilitating and shaping AI-supported learning experiences.

3 Methods

This study employed a qualitative research design using semi-structured interviews to explore interactions between parents, children, and conversational AI tools. The qualitative approach was utilized to gain an in-depth understanding of how families incorporate conversational AI, specifically ChatGPT, into their daily conversations.

3.1 Participants

Parents with children under 13 years who had reported using ChatGPT at home were invited to participate in the study. Participants were recruited using purposive sampling to ensure relevance to the study's objectives. Ethical approval was obtained from the Institutional Review Board (IRB) prior to data collection. Participants were informed about the study's objectives, and written consent was

obtained. They were assured of confidentiality and their right to withdraw at any time without penalty. A total of five parents were interviewed, with four fathers and one mother. Their children's age ranged from 6 to 13 years old, with four boys and one girl represented. Participants reported their adoption of ChatGPT at home ranged from 2 months to over 24 months, and they utilized various devices and modalities including smartphones, tablets, laptops, and combinations of voice and text interactions (see detailed demographics and ChatGPT usage information in Table 1).

3.2 Data Collection & Analysis

The semi-structured interview was designed to comprehensively explore the dynamics of ChatGPT usage in home settings, with a focus on descriptive statistics about usage (e.g., frequency of use, length of adoption) and delve into the content and interaction modes of ChatGPT use. The specific questions included in the interview protocol are as follows:

- When do you first use ChatGPT with your child?
- How often do you use ChatGPT with your child?
- What do you and your child use ChatGPT for?
- How do you and your child use ChatGPT?
- Could you share examples of some typical sessions or interactions using ChatGPT?

The interviews were conducted virtually via the Zoom video conferencing platform to ensure accessibility and convenience for participants. Each session lasted approximately 45–60 minutes. With participants' consent, the interviews were video recorded for subsequent analysis. Three interviews were conducted in English, and two were conducted in Mandarin, based on participants' preferences. Data were securely stored and anonymized during analysis to protect participants' identities.

The interview recordings were transcribed verbatim, and the Mandarin responses were also translated into English for analysis. Thematic analysis was conducted by two authors to identify recurring patterns and key themes related to the use of ChatGPT by parents and children. The thematic analysis followed Braun and Clarke's [6] six-step framework: familiarization with data, generating initial codes, searching for themes, reviewing themes, defining and naming themes, and producing the report.

4 Findings

All five parents reported active use of conversational tools (e.g., ChatGPT, Claude, Doubao) in daily conversations, through both joint use with their children and independent use by the children themselves. All parents reported having prior experience adopting smart home voice assistants (e.g., Siri, Google, and Alexa), but with the advancement of AI technology, particularly LLMs, their preferences has shifted toward Gen-AI tools like ChatGPT, Claude, and Doubao. Through their experiences with ChatGPT, two major themes emerged from the interview analysis: the content of the conversations and the modes of interaction. These themes provide a framework for understanding how AI tools such as ChatGPT have been integrated into everyday interactions in parents and children's lives with diverse contexts and purposes of AI use.

Table 1: Demographics and ChatGPT Usage Information for Five Parents

ID	Parent Role	Child Gender	Child Age	Usage/Month	Device	Modality
P01	Father	Boy	6	24	Smartphone	Voice
P02	Mother	Girl	9	6	Tablet	Text
P03	Father	Boy	13	6	Laptop	Text
P04	Father	Boy	9	2	Laptop	Text
P05	Father	Boy	8	>24	Laptop/Tablet	Voice/Text

4.1 Conversation Content

For the conversation content, parents and children use ChatGPT to involve in many types of conversations. Among the conversations, there are two major themes, depending on the content type: **knowledge-driven** conversations and **play-driven** conversations.

4.1.1 Knowledge-Driven Conversations. Knowledge-driven conversations are those focused on consulting AI tools for either factual information or academic tasks.

Factual Information. Seeking factual information emerged as a prominent theme, with participants reported utilizing AI tools to assist children in gaining accurate understanding of the world. Four out of five participants (P1, P2, P4, P5) reported using ChatGPT for this purpose. The topics explored included general knowledge questions driven by curiosity or everyday wonderings, such as “What is the longest train in the world?” (P1), as well as scientific subjects like biology (P5) and astronomy (P1). For instance, P1 recounted:

“My son asked me about why the core of Mars just stopped. I thought it was a tricky question, and we couldn’t find an answer in any book. So, we asked ChatGPT, and I found the answer to be pretty good... My son is really happy and impressed...but (sometimes) I don’t think the ChatGPT really understands what the question is. After that, he’ll just stop playing with it.” [P1]

This example highlights the content and perceived value of ChatGPT in addressing factual knowledge as an available resources and the child’s reaction to its performance. Parents generally expressed positive attitudes toward ChatGPT’s ability to provide reliable and informative responses. It also indicates that, the interactions between children and ChatGPT can be terminated when the answers are beyond children’s expectations.

Academic Tasks. Academic tasks refer to interactions where AI tools are utilized to support the development of academic skills. These tasks typically involve engaging with ChatGPT to generate, refine, or analyze content aligned with academic objectives, such as curriculum requirements and homework. Four out of five parents (P2, P3, P4, P5) reported either joint or independent use of ChatGPT by their children for academic purposes, including math (P2, P3, P5), writing (P2, P4), and reading (P4). P4 shared a scenario where his child used ChatGPT under his guidance to complete a writing assignment, showcasing ChatGPT’s role in supporting children’s academic learning during their daily conversations:

“The homework was to write a letter based on the scenario where their break time is extended from 10 minutes to 15 minutes and describe what they can do during this time. So, I told him, ‘First, you can ask ChatGPT what things can be done during a 15-minute break.’” [P4]

In this case, the use of ChatGPT for brainstorming writing ideas highlights a distinct purpose compared to seeking factual information, as the child leveraged the tool to explore learning strategies for academic tasks. This example highlights ChatGPT’s promising potential to enhance children’s academic performance through guided and purposeful interaction mediated by parents.

4.1.2 Play-Driven Conversations. Play-driven conversations emphasize the playful and exploratory aspects of interacting with ChatGPT. These interactions often involve utilizing ChatGPT’s creative or entertaining features, such as storytelling (P1, P5), joke-telling (P5), generating artworks (P5), or engaging in casual, open-ended conversations that prioritize curiosity and enjoyment (P1, P2, P5). P1 shared that after seeking factual information, his son often asked “nonsense” or “silly” questions purely for fun. Similarly, P2 recounted her daughter initiated her first conversation with ChatGPT by asking “Who are you?” P5 described his children requesting jokes and asking ChatGPT to generate initial designs for their artwork: “They do like arts, they do like painting, and sometimes they would ask GPT to generate the initial design.” Notably, these conversations sometimes include testing ChatGPT’s abilities and comparing it with other large language models (P2, P5). Two participants (P1, P5) mentioned instances where their children tested ChatGPT’s knowledge as part of their playful engagement. P5 provided an example:

“They feel amazed not only the first time but many times. They were asking which kind of fish is bigger, like a cod or salmon. I think they were trying to have some fun conversations with the AI robot. They had a book (which already had the answers) in front of them. I think they were trying to test its ability.” [P5]

ChatGPT has been used in numerous cases where users consult it for answers to their questions. Users care about how the ChatGPT can correctly answer the questions. However, in some cases where children interact with ChatGPT, they are more interested in playing with ChatGPT by testing its ability to answer questions. This indicates the playful feature of ChatGPT can also be an important factor that attracts children’s engagement. This form of testing reflects a play-driven approach rather than a technical evaluation, where

children derive enjoyment and amazement from exploring ChatGPT's capabilities through playful experimentation, as reported by parents.

4.2 Interaction Mode

Parents reported incorporating ChatGPT into daily conversations through both joint use and children's independent use. In both scenarios, parents engaged in moderation to varying degrees. Moderation involves parental engagements such as teaching, scaffolding, supervising, and monitoring interactions between children and AI tools [8, 18]. Based on the level of parental moderation, we identified two primary modes of interaction: **low-moderation** mode and **high-moderation** mode.

4.2.1 Low-Moderation Mode. In the low-moderation mode, parents adopt a more hands-off approach, allowing children to explore and interact with ChatGPT independently. In this mode, parents typically grant children autonomy, placing minimal restrictions on the time, context, or content of their independent interactions with ChatGPT. Parental moderation is kept to a minimum, with some parents reporting indirect monitoring, such as occasionally checking chat history (P2, P5) or glancing at interactions when passing by (P3). Therefore, children have greater autonomy in their interaction with ChatGPT with minimal parental intervention. In most cases, parents tend to not moderate the conversations strictly. For example, P2 emphasized her goal of fostering independence in her child's use of ChatGPT:

"My goal was to teach her to use it whenever she needs it...I guided her at the beginning in order to teach her how to use it... After that, she started using it on her own...I still check the history occasionally...Because ChatGPT is still text-based, so I feel that with text, there may not be too many distractions...(Interviewer: "So, text is safer than video?") Yeah, it at least is not like short-video platforms which feed a lot of stuff..." [P2]

As a mother of 9-year-old, P2 did not show too much concern about letting her child to use ChatGPT independently. She compared ChatGPT to social media platforms filled with predominantly short-video, which are also prevalent among children recently. She stated that ChatGPT is less distracting or influential, because of its text-based content and its interaction mode is different from short-video platforms that constantly feed videos which can make children addicted to the content. ChatGPT only responds when asked questions and does not provide additional recommendations.

"I'm not worried because...children receive concrete education from official systems such as schools, so using ChatGPT won't easily shape their development." [P4]

Similarly, P4 expressed limited concerns about child use of ChatGPT, as he is confident in the structured education children receive from formal systems such as schools. Parents like P4 may perceive that their children already have a solid educational foundation, which minimizes the likelihood that ChatGPT would significantly shape their overall development. This perspective suggests that parents adopting a low-moderation mode view ChatGPT as a supplementary tool rather than a primary source of education or influence. In this case, ChatGPT is seen as supportive and task-oriented,

serving as a resource for answering questions or assisting with specific activities in daily family conversations.

4.2.2 High-Moderation Mode. On the other hand, in some cases, parents adopt a high-moderation mode, taking an active role in guiding and supporting their children's use of ChatGPT. In this mode, parents reported providing prompts directly to their children (P1, P4), scaffolding follow-up conversations step by step (P1, P4), and testing or clarifying ChatGPT's responses for their children (P1). High-moderation parents reported various reasons for their active engagement. One reason is children's limited proficiency and experience with AI tools. For example, P4 described how he guided his son in refining prompts to generate responses that met their specific needs, emphasizing his role in helping his child navigate ChatGPT effectively.

"He actually doesn't know how to proceed with the current output... So I told him, I said, you can continue your input. You can say, make this 'Catch the tail' to an elementary-level composition. He then type it in following my instructions... I further asked him to ask ChatGPT to revise it to a story (instead of an game instruction), and describe this story." [P4]

Similarly, P5 shared that he often initiates most use cases, explaining that his greater proficiency with ChatGPT enables him to introduce new features to his children:

"I would say I'm mentoring, and guiding a child how to use it... and how you can ask the right question. I think most of the time it's myself (to initiate the use). It's probably I would say 70%." [P5]

Under high moderation mode, parents either feel the need or are willing to take on a moderator role to actively engage their children in interacting with AI tools. This highlights that high parental moderation is often necessary due to children's limited proficiency and experience in navigating AI tools, making parental guidance essential for productive interactions. Another reason for high parental moderation was concern over the potential risks of ChatGPT. P3, a father of a 13-year-old, expressed concerns about the potential harm caused by ChatGPT's responses, particularly in sensitive situations. He stated:

"I guess the main concern is the safety right? ... Some negative or harmful content. Right? Say, the child is depressed. And they use AI as a companion. Then there's a concern of AI being an echo chamber and making the child even more depressed and maybe suicidal." [P3]

Interacting with AI involves the risk of receiving misleading or harmful answers, and as children grow older and their cognitive skills develop, the potential impact of inaccurate or inappropriate guidances may increase. As indicated by P3, parents are particularly worried about ChatGPT's potential influence on children's mental health and its capacity to provide misleading life advice, highlighting the need for careful parental moderation.

5 Discussion

Gen-AI and LLMs like ChatGPT are capable of delivering fast, human-like responses to user queries, making it more easily adopted

by parents and children. However, the integration of AI technology into family conversations raises concerns due to the potential for LLMs to produce irrelevant, false, or misleading information [14, 20]. However, from our study sample, though small, we observed a strong pattern where parents expressed limited concern about using ChatGPT in family conversations.

First, parents consider using ChatGPT as another smart home device which may lower concerns because of its perceived familiarity and functionality. Many parents indicated that they are familiar with conversational agents' functionalities and affordances through their prior experiences using smart speakers, though they reported that these voice assistants are only able to understand simple "voice commands" (e.g., play jokes and music). When using ChatGPT with their children, they prefer the voice mode since it is easier for children to interact with. The familiarity and voice mode together make ChatGPT serve as an alternative to smart speakers like Siri or Alexa, though much more powerful. This perceived understanding of ChatGPT as a more "intelligent" form of voice assistants leads to less concerns about the usage of ChatGPT for parents and children [9]. Future research can dive into this aspect by comparatively investigating the attitudes of parents/children towards smart speakers and ChatGPT and examining how the attitudes toward ChatGPT are influenced by attitudes toward smart speakers. Future design can also take into consideration the potential harms of ChatGPT due to the AI aspects, especially after ChatGPT has been discovered as being used in direct verbal conversations with children.

Second, play-driven conversations in casual settings tend to reduce expectations and concerns about the accuracy of ChatGPT's responses. Almost all parents mention their children "played" with ChatGPT in various ways, like asking random questions (P1, P5), testing its capability (P1, P5), and comparing it with other LLMs (P2). In such scenarios, children perceive ChatGPT as a companion or assistant that can make mistakes [18], rather than a teacher or consultant where accuracy is the primary expectation. As reported in the findings, while parents and children value accurate answers, they also pay attention to how ChatGPT responds and even find it amusing when ChatGPT makes mistakes. Such play-driven interactions not only enhance children's engagement but also deepen their understanding of the underlying mechanisms of LLMs, building their AI literacy through the exploration of ChatGPT's mistakes [18]. Future HCI researchers can focus on this dynamic to design LLMs that are both engaging and effective as learning tools.

Third, an age-related shift in parental concerns about children's use of ChatGPT has been observed. Parents of children under 11 generally express positive attitudes toward their children's free exploration of ChatGPT without active monitoring, while parents of older children raise concerns about the conversation content. This shift may be attributed to age-related cognitive factors. Some parents mentioned that their children's questions to ChatGPT were "easy to answer" (P1), and that children tended to stop the interaction if they did not understand or were dissatisfied with ChatGPT's responses. This indicates that the impact of misleading or incorrect answers may be limited to young children as they are in an early stage of cognitive development, which will be less likely to fully adopt the responses [11, 25]. The text- and voice-based conversation modalities also contribute to the decrease in concerns because they are less intuitive than images or videos for younger children as P2

indicated. However, as children grow older, their social-emotional and cognitive abilities expand yet remain insufficiently mature, making them more susceptible to adopting misleading information or experiencing negative emotions from social interactions [25, 26]. Like the father of a 13-year-old expressed, parents possess concerns about misleading or age-inappropriate content will impact children's physical and mental well-being. These evolving attitudes highlight the need for research on the cognitive and emotional characteristics of child users at different developmental stages to guide the design of age-appropriate AI tools.

6 Conclusion

The study interviewed five parents of children aged 6 to 13, revealing that both children and their parents use ChatGPT in natural home settings for knowledge-driven and play-driven content, either in high-moderation or low-moderation modes. This study contributes to understanding the adoption of AI tools in home settings and provides valuable insights for future design and research on child-centered AI technologies. However, the study is limited by its small sample size due to time constraints, and the interactions between children and ChatGPT were solely reported by parents, lacking input from children themselves or secondary analysis of children's conversation transcripts. Future studies could incorporate multiple sources of qualitative data to enable more comprehensive conversational analysis, examining the diverse content utilized by children across different age groups in the home environment.

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